

Infania Pimentel

Human Factors Engineer · Product Design and UX · Interaction Systems

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Education

TUFTS UNIVERSITY, SCHOOL OF ENGINEERING, Medford, MA

May 2026

Master of Science in Human Factors Engineering

Thesis: Designed and built a wearable device using visual signals to help people connect in person, validated through usability and IRB-approved studies showing increased desire to connect.

UNIVERSITY OF NEW MEXICO, Albuquerque, NM

May 2021

Bachelor of Arts in Chemistry; Bachelor of Science in Psychology

Minor: Computer Science — Cum Laude

Experience

CENTER FOR ENGINEERING EDUCATION AND OUTREACH, Medford, MA

Embedded Product Design Engineer — Componentize Platform

2024–Present

Developed a web platform that enables developers to discover, configure, and connect hardware devices through a browser-based interface.

- Built an ecosystem for connecting sensors, cameras, and microcontrollers to web applications using BLE, WebSocket, and serial communication.
- Designed and embedded an AI assistant into the platform using JavaScript, PyScript, and OpenAI's API.
- Iterated on the design through ongoing developer feedback and prototype testing in Figma and live browsers.

Embedded Systems & Human Factors Engineer — Smart Playground Initiative

2024

Designed and built a screenless interactive toy that teaches kids computational thinking through physical play: a “magic mailbox” that responds when a stuffed animal is placed inside.

- Used signal strength (RSSI) to detect which stuffed animal was placed inside the box and automatically pair with it, replacing screen-based menu selection.
- Built a multi-device system with LED-based feedback to communicate system state (scanning, selection, pairing) without a screen.
- Tested the toy with kids and used their behavior and feedback to refine how the interaction worked.

TUFTS UNIVERSITY, SCHOOL OF ENGINEERING, Medford, MA

Embedded Systems & Human Factors Lead (Master's Thesis) — SociaLIGHT

2024–Present

Developed and evaluated a wearable device that helps people connect in person using social-media-inspired visual signals.

- Prototyped the concept through paper sketches and Figma before building functional hardware end-to-end.
- Switched from BLE/WiFi to NFC to simplify setup, and used visual feedback to communicate system state.
- Designed and ran a two-part study (usability testing and IRB-approved behavioral study); refined the design across iterations based on findings, cutting setup time by 83% and reaching 100% task completion.

TUFTS UNIVERSITY, ENP 110 — HUMAN FACTORS IN MEDICAL TECHNOLOGY, Medford, MA

Medical Technology Consulting Practicum — Michael Wiklund, Instructor

2025

Completed six medical device HFE projects spanning usability testing, UI requirements, and regulatory-compliant warning and alarm design.

- Completed a four-month anatomy and cadaver lab to ground design decisions in real clinical context.
- Conducted formative usability tests of a glucose meter, producing a full test protocol and design recommendations.
- Designed alarms and warnings for medical devices per IEC 60601-1-8 and ANSI Z535 standards.

BRAY LABS MACHINE SHOP, Medford, MA

Machine Shop Fabrication Technician

2023–2024

- Trained students on subtractive and additive manufacturing equipment and advised on manufacturability and design constraints.

UNIVERSITY OF NEW MEXICO, NEUROSCIENCES, Albuquerque, NM

Research Technician

2021–2023

Conducted EEG and fNIRS research and developed MATLAB/R algorithms modeling hippocampal dopamine activity during novelty and movement.

Skills

Product Design: Figma, Adobe Suite, Wireframing, Prototyping, Interaction Design, Onshape, SolidWorks

AI & Software: OpenAI API, Python, JavaScript, PyScript, HTML/CSS, MATLAB, R

Embedded Systems & Hardware: ESP32, Raspberry Pi, BLE, NFC, Sensors, Motors, Cameras, LED Systems

Human Factors & Research: Usability Testing, IRB Study Design, Ergonomic Analysis, Statistical Analysis, Risk Analysis, IEC 62366-1, FDA HFE Guidance

Fabrication: Laser Cutting, Waterjet, Additive Manufacturing, DFM, Milling

Languages: English (Fluent), Spanish (Reading/Writing)